

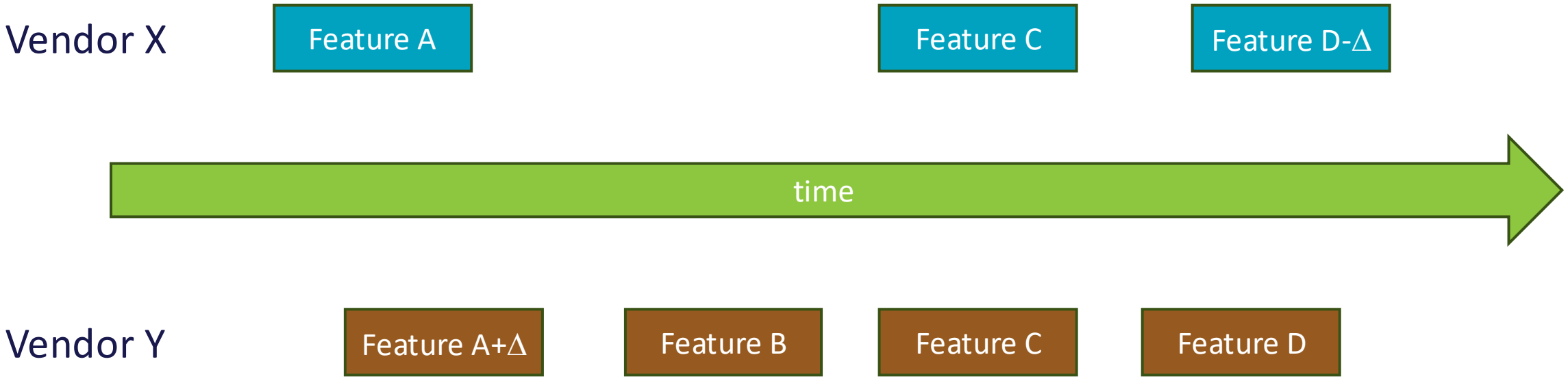
x86 Future Direction

Collaborative approach to accelerate AI-era infrastructure ecosystem deployment

Compatibility. Predictability. Developer confidence.

A joint update from AMD + Intel

Past Approach to x86 Feature Development



Prior x86 Divergence → Problem Statement

Ecosystem

Extra development and validation effort for unique feature code paths

or

Least-common-denominator feature adoption

x86 CPU Vendors

Wasted silicon die area, power, and R&D effort for unadopted features

and

Ecosystem frustration

x86 Ecosystem Advisory Group

Luminaries



Tim Sweeney



Linus Torvalds

BROADCOM · DELL Technologies · Google · Hewlett Packard Enterprise · hp · NUTANIX · Adobe · Lenovo · Meta · Microsoft · ORACLE · Red Hat

The New Way Forward

Adopt Open Compute philosophy…

1. Jointly engage ecosystem on future workload requirements
2. Collaborate on features and specs to address requirements
3. Compete vigorously on implementation

Outcomes from First Two Years of the “New Way”

AVX10

Portable vector instructions

Consistent vector code across x86

Advanced Vector Extensions

ACE

Matrix multiplication acceleration

Built-in AI math acceleration on x86 CPUs

AI Compute Extensions

APX

Increased registers & ISA features

Improved general-purpose code performance

Advanced Performance Extensions

FRED

Modernized interrupt model

Faster, more reliable servers

Flexible Return & Event Delivery

MTT

Unified memory tagging

Enhanced memory-safety for accidental and malicious attacks

Memory Tagging Technology

Panel Discussion

Moderators



Robert Hormuth
CVP, Architecture and Strategy
AMD



Jeff McVeigh
VP, Data Center Strategic Programs
Intel

Panelists



Martin Dixon
Engineering Director
Google



Andrew Wheeler
Deputy CTO
SVP & Director of HPE Labs
HPE



Jon Lange
Distinguished Engineer
Microsoft



Steve Watt
VP, Distinguished Engineer
Research & Emerging Tech
RedHat

Public

Thank You!



OCP GLOBAL SUMMIT

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